

Data Flow Diagram

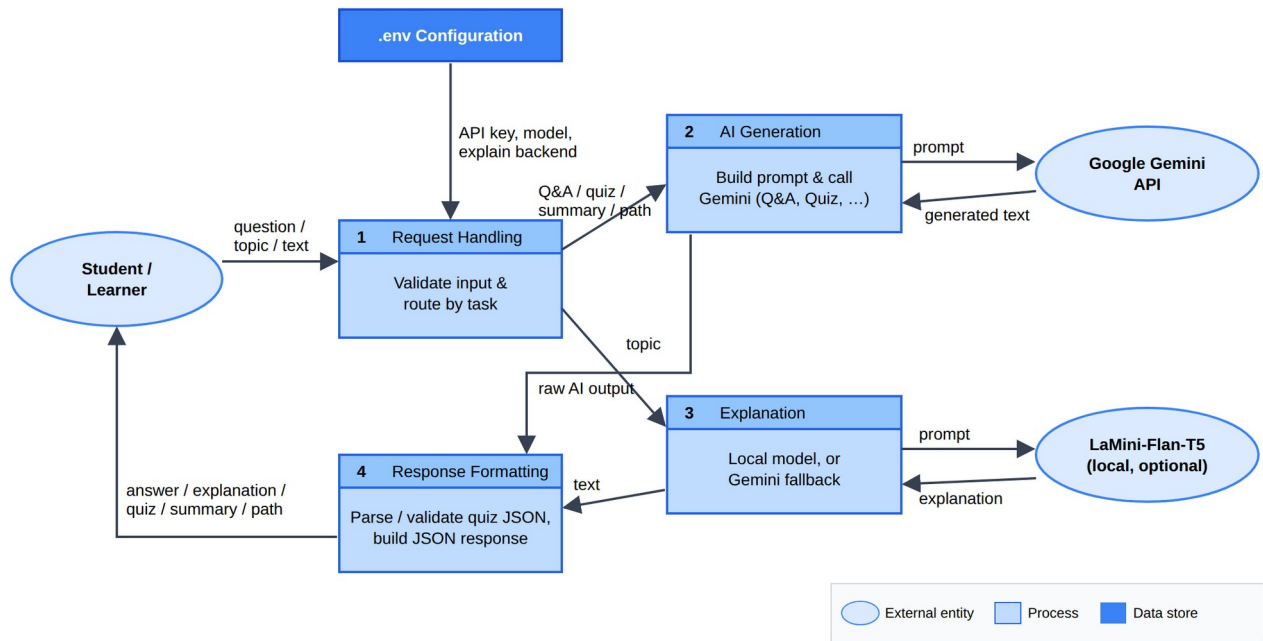
Date	03 October 2026
Team ID	SWTID-2026-7859
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	2 Marks

Data Flow Diagram

The Data Flow Diagram (DFD) below shows how data moves through EduGenie - between external entities, processes and data stores.

DFD Symbol Legend

Symbol	Name	Description (in EduGenie)
Oval / Rounded shape	External Entity	Student / Learner, Google Gemini API, LaMini-Flan-T5 local model
Rectangle with numbered header	Process	1 Request Handling, 2 AI Generation, 3 Explanation, 4 Response Formatting
Rectangle (solid fill, no number)	Data Store	.env configuration (API key, model name, explanation backend). EduGenie stores no user data.
Labeled arrow	Data Flow	Question / topic / text, prompts, generated text, JSON responses

EduGenie – Data Flow Diagram (Level 1)

Figure: EduGenie Data Flow Diagram (Level 1)

Process Description

Process	Input	Output	Description
1. Request Handling	Question, topic or passage from the student	Validated request routed to the right module	FastAPI endpoints (/qa, /explain, /quiz, /summarize, /learn/recommendations) check for empty / too-long input (HTTP 400).
2. AI Generation	Validated text + feature-specific prompt	Raw Gemini text / JSON	gemini_client.py sends the prompt to Google Gemini (gemini-flash-latest) using the API key from .env.
3. Explanation	Topic	Simple explanation	Uses the local LaMini-Flan-T5-783M model when installed, otherwise (or on error) falls back to Gemini.
4. Response Formatting	Raw AI output	JSON response for the browser	Cleans Markdown code fences, validates quiz JSON (4 options, answer in options) and maps errors to 502/503.